Assumptions:

$$m = 0.856\ kg \rightarrow Total\ vehicle\ mass$$

$$v_{\text{top}} = 17.4\ mph \rightarrow 7.78\frac{m}{s} \rightarrow Top\ Speed$$

$$v_{60A} = 4.17\ mph \rightarrow 1.864\frac{m}{s} \rightarrow Speed\ @\ 60A\ and\ 5000RPM$$

$$F_t = 3.54\ \text{N} \rightarrow \text{Tractive Force}$$

$$R_r = 0.053\ \text{N} \rightarrow\ Rolling\ Resistance$$

$$a_{\max} = 4.07\frac{m}{s^2} \rightarrow Max\ Acceleration$$

$$E_{PLA} = 3.0\ GPa\ (Assumed\ Young's\ Modulus)$$

$$b = 3.00\ \text{in} \rightarrow Chassis\ Width$$

$$t = 0.25\ \text{in} \rightarrow Thickness$$

$$L_b = 1.60\ \text{in} \rightarrow 0.04064\ m \rightarrow Length\ of\ deformation$$

$$A = bt \rightarrow 0.75\ in^2 \rightarrow Chassis\ cross-sectional\ area \rightarrow 4.839 * 10^{-4} m^2$$

$$k = \frac{AE}{L_b} = \frac{(4.839 * 10^{-4})(3.0 * 10^9)}{0.04064} \rightarrow Axial\ Chassis\ Stiffness \rightarrow 3.57 * 10^7 N/m$$

Impact force at top speed:

$\frac{1}{2}mv^2 = \frac{1}{2}k\delta^2$ Elastic-impact approximation, since $F_{max} = k\delta$

$$F_{max} = v\sqrt{mk} \rightarrow Using\ top\ speed \rightarrow F_I = 4.30 * 10^4 N\ or\ 43.008\ kN$$

Peak force predicted by the simplified linear-elastic model.

Chassis Deflection at top speed:

$$\delta_c = \frac{F_I}{k} \rightarrow \frac{43{,}008}{3.57 * 10^7} = 1.204\ mm$$

Chassis Stress at top speed:

$$\sigma_c = \frac{F_I}{A} \rightarrow \frac{43{,}008}{4.839 * 10^{-4}} \rightarrow 8.89 * 10^7 Pa \rightarrow 88.9\ MPa$$

Predicted stress not accounting for filament type, print orientation, infill, or wall count of PLA.

Lower-speed impact at the 60-A operating point:

$$F_I = 1.864\sqrt{(0.856)(3.57 * 10^7)} = 1.0304 * 10^4 N \rightarrow 10.304\ kN$$

$$\delta_c = \frac{10{,}304}{3.57 \times 10^7} = 0.0002886 \rightarrow 0.289\ mm$$

$$\sigma_c = \frac{10{,}304}{4.839 \times 10^{-4}} = 2.13 * 10^7\ Pa \rightarrow 21.3\ MPa$$

Impact calculation assumes the kinetic energy is absorbed elastically by the simplified chassis section. These simplified calculations do not account for tires, mounts, printed PLA deformation/fracture, vehicle rotation, and contact geometry absorbing energy and changing the actual peak force.

Hypothetical Axle model:

$$d_a = 5\ mm \rightarrow 0.005\ m\ (Assumed)$$

$$L_a = 12.7\ mm \rightarrow 0.0127\ m\ (Assumed)$$

$$E_{steel} = 200\ GPa(Assumed\ Young's\ Modulus)$$

Symmetric wheel-first collision:

$$F_w = \frac{F_I}{2}\ (Impact\ force\ carried\ by\ one \frac{wheel}{axle} end)$$

Top Speed:

$$F_w = \frac{43008}{2} = 21504\ N \rightarrow 21.50\ kN$$

Axle bending moment:

$$M = F_w L_a \rightarrow (21504)(0.0127) = 273.1\ N * m$$

Axle Stress:

$$\sigma_a = \frac{32M}{\pi d^3}\ (Solid\ Circular\ axle) \rightarrow \frac{32(273.1)}{\pi(0.005)^3} \rightarrow 2.23 * 10^{10} Pa = 22.3\ GPa$$

Axle Deflection:

$$\delta_a = \frac{F_w L_a^3}{3EI}\ (Cantilever)$$

$$I = \frac{\pi d^4}{64}\ (Solid\ round\ axle) \rightarrow \frac{\pi(0.005)^4}{64} = 3.068 * 10^{-11} m^4$$

$$\delta_a = \frac{F_w L_a^3}{3EI} \rightarrow \frac{21504(0.0127)^3}{3(200 * 100^9)(3.068 * 10^{-11})} = 0.00239\ m\ or\ 2.39\ mm$$

Lower Speed:

$$F_w = \frac{10.304}{2} = 5.152\ kN$$

Moment:

$$M = (5152)(0.0127) = 65.43\ N * m$$

Stress:

$$\sigma_a = \frac{32(65.43)}{\pi(0.005)^3} = 5.33 * 10^9 Pa\ or\ 5.33\ GPa$$

Deflection:

$$\delta_a = \frac{F_w L_a^3}{3EI} \rightarrow \frac{5152(0.0127)^3}{3(200 * 100^9)(3.068 * 10^{-11})} = 0.000573\ m\ or\ 0.573\ mm$$

The calculated axle stress greatly exceeds typical steel yield strengths. Therefore, these results indicate predicted axle failure under the assumed wheel-first impact load. The calculated elastic axle deflections should be interpreted only as preliminary linear-elastic values, actual deformation after yielding would be nonlinear.